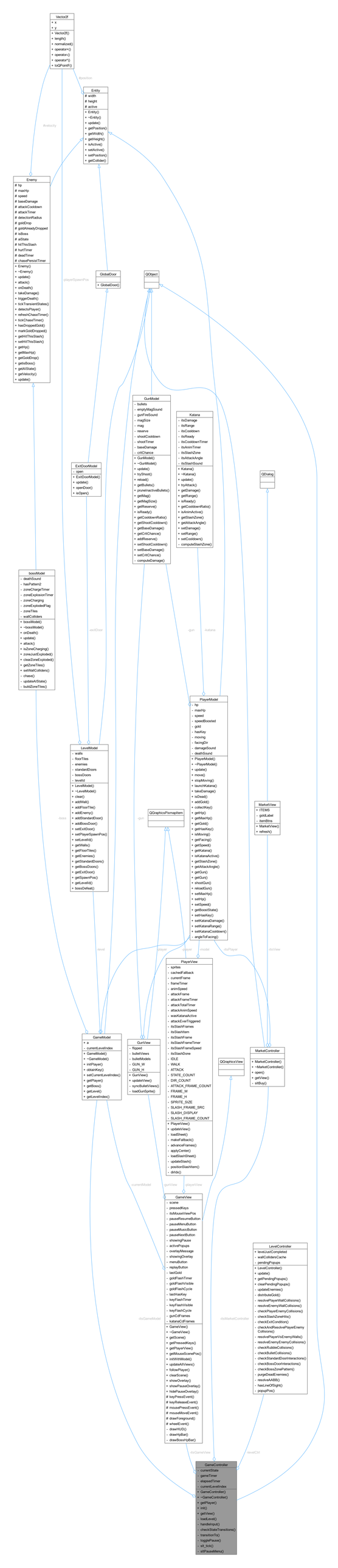

Vector2f
+ x
+ y
+ Vector2f()
+ length()
+ normalized()
+ operator+()
+ operator-()
+ operator*()
+ toQPointF()
#position
#velocity
Entity
# width
# height
# active
+ Entity()
+ ~Entity()
+ update()
+ getPosition()
+ getWidth()
+ getHeight()
+ isActive()
+ setActive()
+ setPosition()
+ getCollider()
Enemy
# hp
# maxHp
# speed
# baseDamage
# attackCooldown
# attackTimer
# detectionRadius
# goldDrop
# goldAlreadyDropped
# isBoss
# aiState
# hitThisSlash
# hurtTimer
# deadTimer
# chasePersistTimer
+ Enemy()
+ ~Enemy()
+ update()
+ attack()
+ onDeath()
+ takeDamage()
+ triggerDeath()
+ tickTransientStates()
+ detectsPlayer()
+ refreshChaseTimer()
+ tickChaseTimer()
+ hasDroppedGold()
+ markGoldDropped()
+ getHitThisSlash()
+ setHitThisSlash()
+ getHp()
+ getMaxHp()
+ getGoldDrop()
+ getisBoss()
+ getAIState()
+ getVelocity()
+ update()
-playerSpawnPos
GlobalDoor
+ GlobalDoor()
QObject
GunModel
- bullets
- emptyMagSound
- gunFireSound
- magSize
- mag
- reserve
- shootCooldown
- shootTimer
- baseDamage
- critChance
+ GunModel()
+ ~GunModel()
+ update()
+ tryShoot()
+ reload()
+ getBullets()
+ pruneInactiveBullets()
+ getMag()
+ getMagSize()
+ getReserve()
+ isReady()
+ getCooldownRatio()
+ getShootCooldown()
+ getBaseDamage()
+ getCritChance()
+ addReserve()
+ setShootCooldown()
+ setBaseDamage()
+ setCritChance()
- computeDamage()
Katana
- itsDamage
- itsRange
- itsCooldown
- itsReady
- itsCooldownTimer
- itsAnimTimer
- itsSlashZone
- itsAttackAngle
- itsSlashSound
+ Katana()
+ ~Katana()
+ update()
+ tryAttack()
+ getDamage()
+ getRange()
+ isReady()
+ getCooldownRatio()
+ isAnimActive()
+ getSlashZone()
+ getAttackAngle()
+ setDamage()
+ setRange()
+ setCooldown()
- computeSlashZone()
QDialog
ExitDoorModel
- open
+ ExitDoorModel()
+ update()
+ openDoor()
+ isOpen()
bossModel
- deathSound
- hasPattern2
- zoneChargeTimer
- zoneExplosionTimer
- zoneCharging
- zoneExplodedFlag
- zoneTiles
- wallColliders
+ bossModel()
+ ~bossModel()
+ onDeath()
+ update()
+ attack()
+ isZoneCharging()
+ zoneJustExploded()
+ clearZoneExploded()
+ getZoneTiles()
+ setWallColliders()
- chase()
- updateAIState()
- buildZoneTiles()
-exitDoor
-gun
-katana
PlayerModel
- hp
- maxHp
- speed
- speedBoosted
- gold
- hasKey
- moving
- facingDir
- damageSound
- deathSound
+ PlayerModel()
+ ~PlayerModel()
+ update()
+ move()
+ stopMoving()
+ launchKatana()
+ takeDamage()
+ isDead()
+ addGold()
+ collectKey()
+ getHp()
+ getMaxHp()
+ getGold()
+ getHasKey()
+ isMoving()
+ getFacing()
+ getSpeed()
+ getKatana()
+ isKatanaActive()
+ getSlashZone()
+ getAttackAngle()
+ getGun()
+ getGun()
+ shootGun()
+ reloadGun()
+ setMaxHp()
+ setHp()
+ setSpeed()
+ getBoostState()
+ setHasKey()
+ setKatanaDamage()
+ setKatanaRange()
+ setKatanaCooldown()
- angleToFacing()
LevelModel
- walls
- floorTiles
- enemies
- standardDoors
- bossDoors
- levelId
+ LevelModel()
+ ~LevelModel()
+ clear()
+ addWall()
+ addFloorTile()
+ addEnemy()
+ addStandardDoor()
+ addBossDoor()
+ setExitDoor()
+ setPlayerSpawnPos()
+ setLevelId()
+ getWalls()
+ getFloorTiles()
+ getEnemies()
+ getStandardDoors()
+ getBossDoors()
+ getExitDoor()
+ getSpawnPos()
+ getLevelId()
+ bossDefeat()
-boss
-gun
QGraphicsPixmapItem
MarketView
+ ITEMS
- goldLabel
- itemBtns
+ MarketView()
+ refresh()
-level
-player
-player
-model
-itsPlayer
-itsView
PlayerView
- sprites
- cachedFallback
- currentFrame
- frameTimer
- animSpeed
- attackFrame
- attackFrameTimer
- attackTotalTimer
- attackAnimSpeed
- wasKatanaActive
- attackEverTriggered
- itsSlashFrames
- itsSlashItem
- itsSlashFrame
- itsSlashFrameTimer
- itsSlashFrameSpeed
- itsSlashDone
- IDLE
- WALK
- ATTACK
- STATE_COUNT
- DIR_COUNT
- ATTACK_FRAME_COUNT
- FRAME_W
- FRAME_H
- SPRITE_SIZE
- SLASH_FRAME_SRC
- SLASH_DISPLAY
- SLASH_FRAME_COUNT
+ PlayerView()
+ updateView()
- loadSheet()
- makeFallback()
- advanceFrames()
- applyCenter()
- loadSlashSheet()
- updateSlash()
- positionSlashItem()
- dirIdx()
GameModel
+ a
- currentLevelIndex
+ GameModel()
+ ~GameModel()
+ initPlayer()
+ obtainKey()
+ setCurrentLevelIndex()
+ getPlayer()
+ getBoss()
+ getLevel()
+ getLevelIndex()
GunView
- flipped
- bulletViews
- bulletModels
- GUN_W
- GUN_H
+ GunView()
+ updateView()
- syncBulletViews()
- loadGunSprite()
QGraphicsView
MarketController
+ MarketController()
+ ~MarketController()
+ open()
+ getView()
- sltBuy()
-currentModel
-gunView
-playerView
GameView
- scene
- pressedKeys
- itsMouseViewPos
- pauseResumeButton
- pauseMenuButton
- pauseMusicButton
- pauseNextButton
- showingPause
- activePopups
- overlayMessage
- showingOverlay
- menuButton
- replayButton
- lastGold
- goldFlashTimer
- goldFlashVisible
- goldFlashCycle
- lastHasKey
- keyFlashTimer
- keyFlashVisible
- keyFlashCycle
- gunCdFrames
- katanaCdFrames
+ GameView()
+ ~GameView()
+ getScene()
+ getPressedKeys()
+ getPlayerView()
+ getMouseScenePos()
+ initWithModel()
+ updateAllViews()
+ followPlayer()
+ clearScene()
+ showOverlay()
+ showPauseOverlay()
+ hidePauseOverlay()
# keyPressEvent()
# keyReleaseEvent()
# mousePressEvent()
# mouseMoveEvent()
# drawForeground()
# wheelEvent()
- drawHUD()
- drawHpBar()
- drawBossHpBar()
LevelController
+ levelJustCompleted
- wallCollidersCache
- pendingPopups
+ LevelController()
+ update()
+ getPendingPopups()
+ clearPendingPopups()
- updateEnemies()
- distributeGold()
- resolvePlayerWallCollisions()
- resolveEnemyWallCollisions()
- checkPlayerEnemyCollisions()
- checkSlashZoneHits()
- checkExitCondition()
- checkAndResolvePlayerEnemy Collisions()
- resolvePlayerVsEnemyWalls()
- resolveEnemyEnemyCollisions()
- checkRubbleCollisions()
- checkBulletCollisions()
- checkStandardDoorInteractions()
- checkBossDoorInteractions()
- checkBossZonePattern()
- purgeDeadEnemies()
- resolveAABB()
- hasLineOfSight()
- popupPos()
-itsGameModel
-itsMarketController
-itsGameView
-levelCtrl
GameController
- currentState
- gameTimer
- elapsedTimer
- currentLevelIndex
+ GameController()
+ ~GameController()
+ getPlayer()
+ init()
+ getView()
- loadLevel()
- handleInput()
- checkStateTransitions()
- transitionTo()
- togglePause()
- slt_tick()
- sltPauseMenu()